Day 3: Introduction to IP Addressing Theory

Task 1 : List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

IPv4 and IPv6 are two versions of the Internet Protocol used to identify devices and route data across networks. IPv6 was developed to address the shortage of IPv4 addresses and provide improvements in network communication.

IPv4	IPv6
Uses a 32-bit address.	Uses a 128-bit address.
Written in dotted-decimal format, e.g., 192.168.1.10.	Written in hexadecimal groups separated by colons, e.g., 2001:db8::10.
Provides approximately 4.3 billion total addresses.	Provides an extremely large address space of 2^{128} addresses.
Uses subnet masks such as 255.255.255.0 or CIDR notation such as /24.	Uses prefix lengths such as /64 to identify the network portion.
Supports broadcast communication.	Does not use broadcast; it uses multicast and anycast instead.
Often uses NAT to allow multiple private devices to share a public IPv4 address.	Supports a much larger address space, reducing the need for address-sharing NAT, although NAT-like techniques can still be used.
Address configuration can be manual or use DHCP.	Supports manual configuration, SLAAC, and DHCPv6.
IP headers have a variable length, typically 20–60 bytes without options.	The base IP header has a fixed length of 40 bytes.

Importance of IPv4 and IPv6

IPv4 remains widely used in home, office, and enterprise networks. IPv6 provides a much larger address space and modern features that help support the growing number of internet-connected devices.

Task 2 : Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Step 1: Network Details

- Network Address: 192.168.10.0
- Subnet Mask: 255.255.255.0 (/24)
- Default Gateway: 192.168.10.1
- Broadcast Address: 192.168.10.255
- Valid Host Range: 192.168.10.1 – 192.168.10.254

Step 2: IP Address Allocation Table

No.	Device	IP Address	Subnet Mask	Default Gateway
1	Laptop	192.168.10.10	255.255.255.0	192.168.10.1
2	Smartphone	192.168.10.11	255.255.255.0	192.168.10.1
3	Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
4	Printer	192.168.10.13	255.255.255.0	192.168.10.1

Step 3: Explanation

Each device is assigned a unique IP address within the same subnet, preventing address conflicts. The subnet mask 255.255.255.0 identifies the local network, while the default gateway 192.168.10.1 allows devices to communicate with networks outside the local subnet.

Step 4: Connectivity Verification

After assigning the addresses, connectivity can be checked using the following commands:

- ipconfig — verifies the IP address, subnet mask, and gateway on Windows.
- ping 192.168.10.1 — tests connectivity to the default gateway.
- ping 192.168.10.13 — from another device, tests connectivity to the printer if ping is enabled and the devices can communicate directly.

Task 3 : Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

Step 1: Identify the Network Details

- Given Network: 192.168.1.0/24
- New CIDR Prefix: /27
- Subnet Mask: 255.255.255.224

Note: The question mentions a /27 subnet mask on a 192.168.1.0 network. This calculation assumes the original network is 192.168.1.0/24.

Step 2: Calculate the Number of Subnets

Original prefix = /24
New prefix = /27

Borrowed bits = 27 − 24 = 3

Number of subnets = 2^3 = **8 subnets**

Step 3: Calculate Usable Hosts per Subnet

IPv4 has 32 bits.

Host bits = 32 − 27 = 5

Total addresses per subnet = 2^5 = 32

Usable hosts per subnet = 32 − 2 = **30 hosts**

Two addresses are reserved for the network address and broadcast address.

Step 4: Calculate the Block Size

Subnet mask: 255.255.255.224

Block size = 256 − 224 = **32**

Therefore, subnet network addresses increase in increments of 32 in the last octet.

Step 5: First Three Subnet Ranges

No.	Network Address	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.1.0/27	192.168.1.1	192.168.1.30	192.168.1.31
2	192.168.1.32/27	192.168.1.33	192.168.1.62	192.168.1.63
3	192.168.1.64/27	192.168.1.65	192.168.1.94	192.168.1.95

Final Answers

- **Subnet Mask:** 255.255.255.224
- **Total Subnets:** 8
- **Usable Hosts per Subnet:** 30
- **Block Size:** 32
- **First Three Network Addresses:** 192.168.1.0, 192.168.1.32, and 192.168.1.64

Task 4 : You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Given Information

- IP Address: 10.0.5.17
- Subnet Mask: 255.255.255.0

Step 1: Convert Subnet Mask to CIDR Notation

Subnet Mask: 255.255.255.0

Binary representation:

11111111.11111111.11111111.00000000

The mask contains 24 network bits.

Therefore, the CIDR notation is **/24**.

Step 2: Calculate the Network Address Using AND Operation

IP Address: 10.0.5.17

Subnet Mask: 255.255.255.0

Binary representation of the last octet:

IP Address: 00010001
Subnet Mask: 00000000

Performing the AND operation:

00010001 AND 00000000 = 00000000

Therefore, the network address is **10.0.5.0**.

Step 3: Calculate the Broadcast Address

For a /24 network, all 8 host bits are set to 1.

Last octet in binary: 11111111

11111111 in decimal = 255

Therefore, the broadcast address is **10.0.5.255**.

Step 4: Calculate the Valid Host Range

The first valid host address is the network address plus 1:

First Valid Host: **10.0.5.1**

The last valid host address is the broadcast address minus 1:

Last Valid Host: **10.0.5.254**

Step 5: Calculate the Number of Usable Hosts

Host bits = 32 − 24 = 8

Total IP addresses = 2^8 = 256

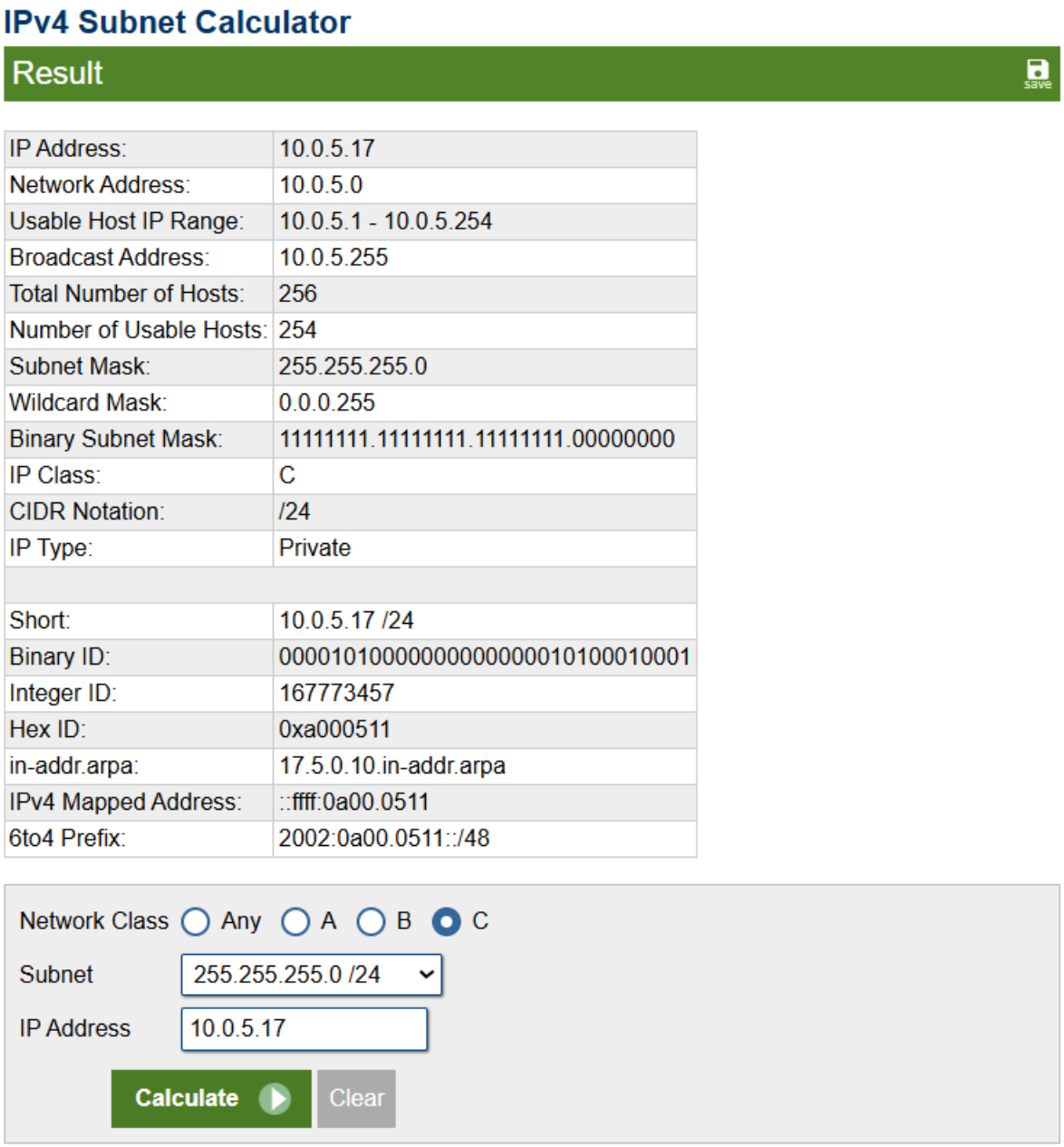
Usable hosts = 256 − 2 = **254**

The network and broadcast addresses cannot be assigned to ordinary hosts.

Final Answer

Parameter	**Result**
Given IP Address	10.0.5.17
Subnet Mask	255.255.255.0
CIDR Notation	/24
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
First Valid Host	10.0.5.1
Last Valid Host	10.0.5.254
Usable Hosts	254

Task 5 : Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.



IPv4 Subnet Calculator

Result

IP Address:	10.0.5.17
Network Address:	10.0.5.0
Usable Host IP Range:	10.0.5.1 - 10.0.5.254
Broadcast Address:	10.0.5.255
Total Number of Hosts:	256
Number of Usable Hosts:	254
Subnet Mask:	255.255.255.0
Wildcard Mask:	0.0.0.255
Binary Subnet Mask:	11111111.11111111.11111111.00000000
IP Class:	C
CIDR Notation:	/24
IP Type:	Private
Short:	10.0.5.17 /24
Binary ID:	00001010000000000000010100010001
Integer ID:	167773457
Hex ID:	0xa000511
in-addr.arpa:	17.5.0.10.in-addr.arpa
IPv4 Mapped Address:	::ffff:0a00.0511
6to4 Prefix:	2002:0a00.0511::/48

Network Class ○ Any ○ A ○ B ● C

Subnet 255.255.255.0 /24

IP Address 10.0.5.17

Calculate Clear

The results from the online subnet calculator matched my manual calculations. The network address is 10.0.5.0, the broadcast address is 10.0.5.255, and the valid host range is 10.0.5.1 to 10.0.5.254.